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◦ • Finance Credit Follow-
Up Email Agent
AI Enablement
Internship – Task 2

Submitted By:

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Problem Statement

- Challenges in Finance Follow-Ups
- Manual payment reminders are repetitive
- Delayed communication increases overdue payments
- Tone inconsistency affects client relationships
- Lack of audit tracking
- Manual escalation handling is inefficient
- Objective
- Build an automated AI-powered reminder system.

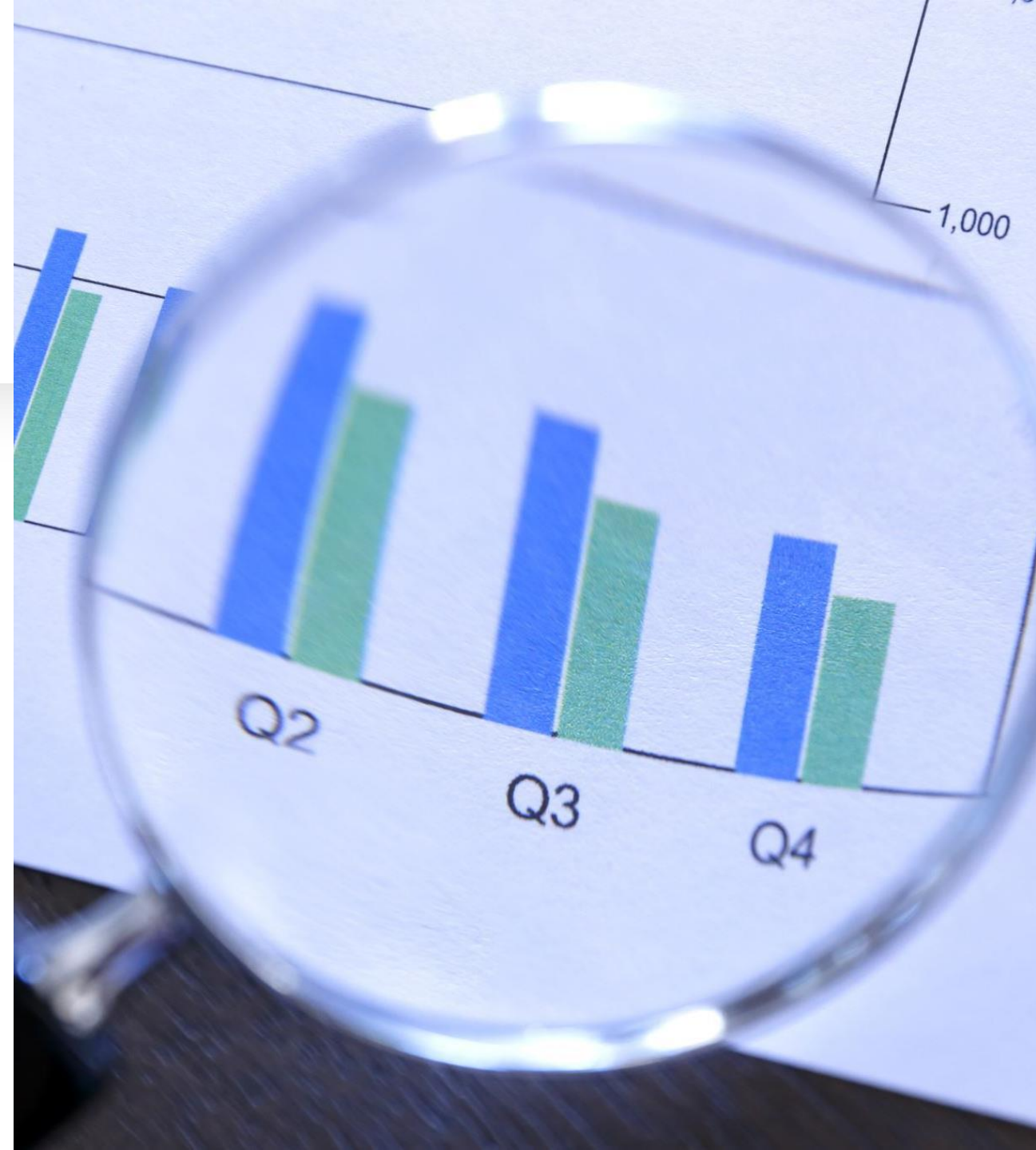


Proposed Solution

Solution Overview

The system:

- Reads invoice data from CSV
- Calculates overdue days
- Applies escalation logic
- Generates personalized emails
- Stores audit logs
- Flags legal escalation cases



System Architecture

- CSV Data
- ↓
- Pandas Processing
- ↓
- Escalation Engine
- ↓
- Email Generator
- ↓
- Audit Logs
- ↓
- Streamlit Dashboard



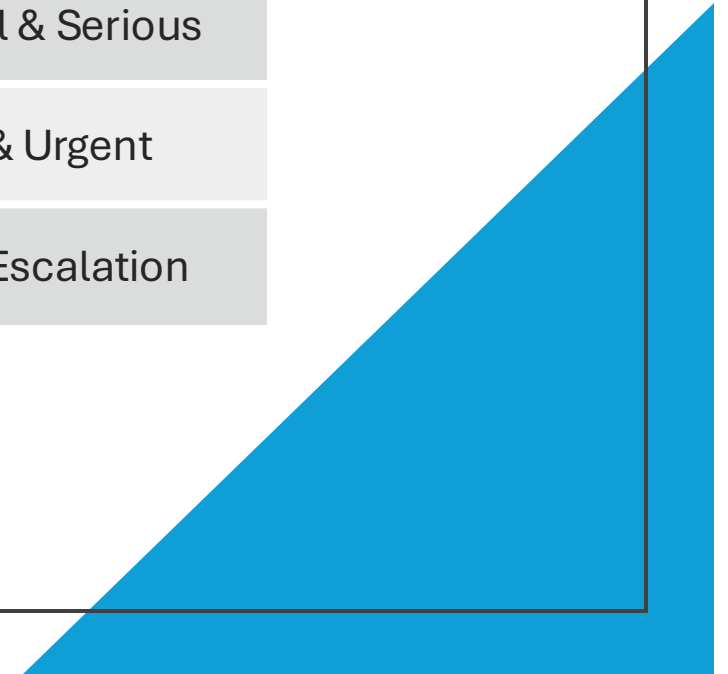
Technologies Used



Component	Technology
Language	Python
UI	Streamlit
Data	Pandas
Logs	JSON
Config	dotenv

Tone Escalation Logic

Overdue Days	Tone
1–7	Warm & Friendly
8–14	Polite but Firm
15–21	Formal & Serious
22–30	Stern & Urgent
30+	Legal Escalation



Results & Sample Outputs

Dashboard Overview

Finance Email Agent

Ask Gemini

localhost:8501

GmailNew TabNew TabGmailYouTubeMapsTranslateNewsPhysics WallahPrime Video: SearchMy coursesStep64Not Passed

File change.RerunAlways rerun

 **Finance Credit Follow-Up Email Agent**

AI-powered invoice reminder and escalation system

Invoice Records



	invoice_no	client_name	amount	due_date	email	followup_count
0	INV001	Rajesh Kapoor	45000	2026-05-05	rajesh@gmail.com	1
1	INV002	Priya Sharma	12000	2026-04-28	priya@gmail.com	2
2	INV003	Amit Verma	78000	2026-04-20	amit@gmail.com	3
3	INV004	TechNova Pvt Ltd	150000	2026-04-10	finance@technova.com	4

Generate Emails

 **Audit Logs**

```
{"client": "Rajesh Kapoor", "invoice": "INV001", "tone": "Warm & Friendly", "status": "Dry Run", "timestamp": "2026-05-08 23:33:18.526945"}
```

```
{"client": "Priya Sharma", "invoice": "INV002", "tone": "Polite but Firm", "status": "Dry Run", "timestamp": "2026-05-08 23:33:18.527947"}
```

```
...
```

Audit Trail Logs

Finance Email Agent

Ask Gemini

localhost:8501

GmailNew TabNew TabGmailYouTubeMapsTranslateNewsPhysics WallahPrime Video: SearchMy coursesStep64Not Passed

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0	INV001	Rajesh Kapoor	45000	2026-05-05	rajesh@gmail.com	1
1	INV002	Priya Sharma	12000	2026-04-28	priya@gmail.com	2
2	INV003	Amit Verma	78000	2026-04-20	amit@gmail.com	3
3	INV004	TechNova Pvt Ltd	150000	2026-04-10	finance@technova.com	4

Generate Emails

Audit Logs

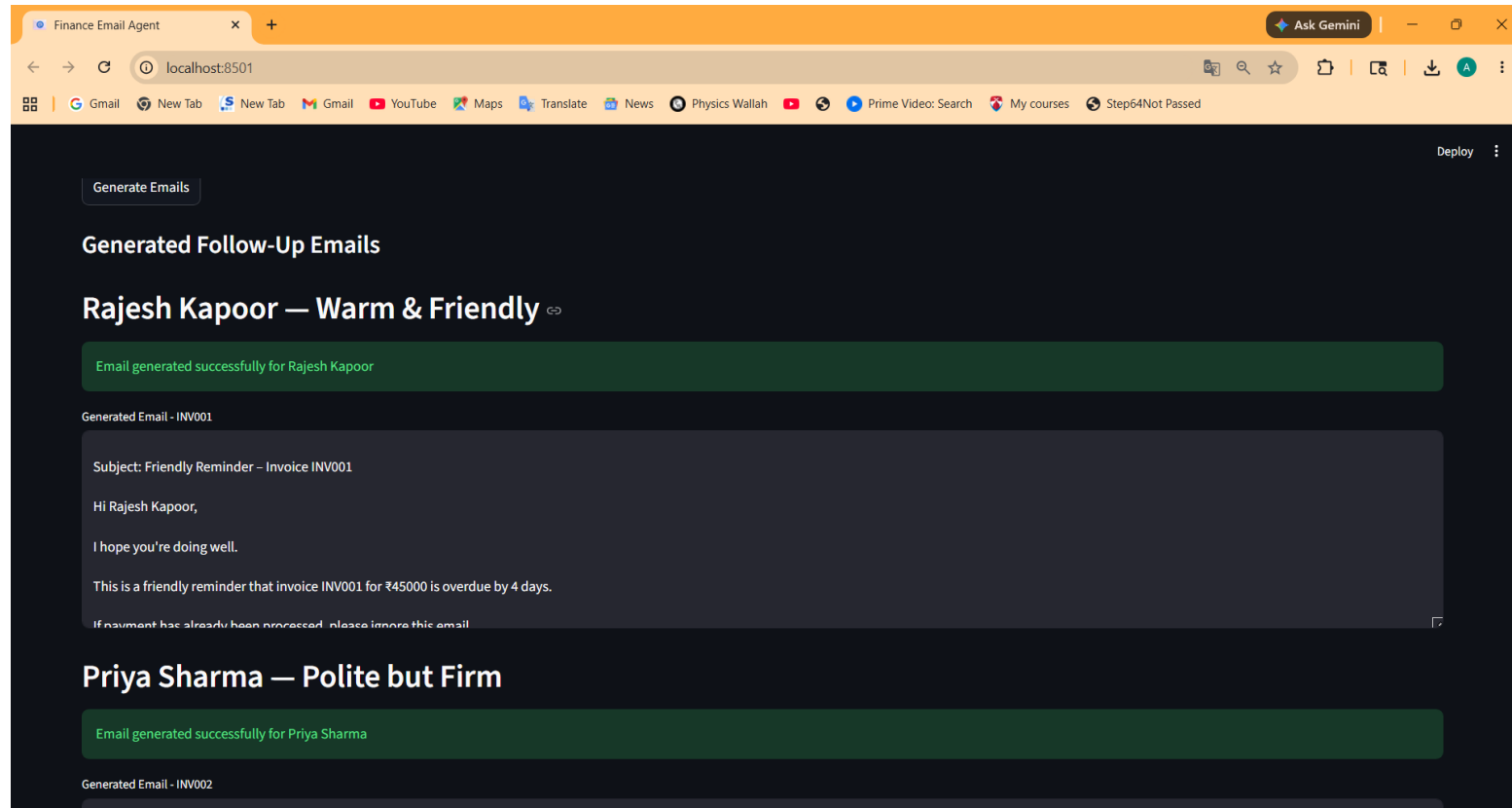
```
{ "client": "Rajesh Kapoor", "invoice": "INV001", "tone": "Warm & Friendly", "status": "Dry Run", "timestamp": "2026-05-08 23:33:18.526945" }
```

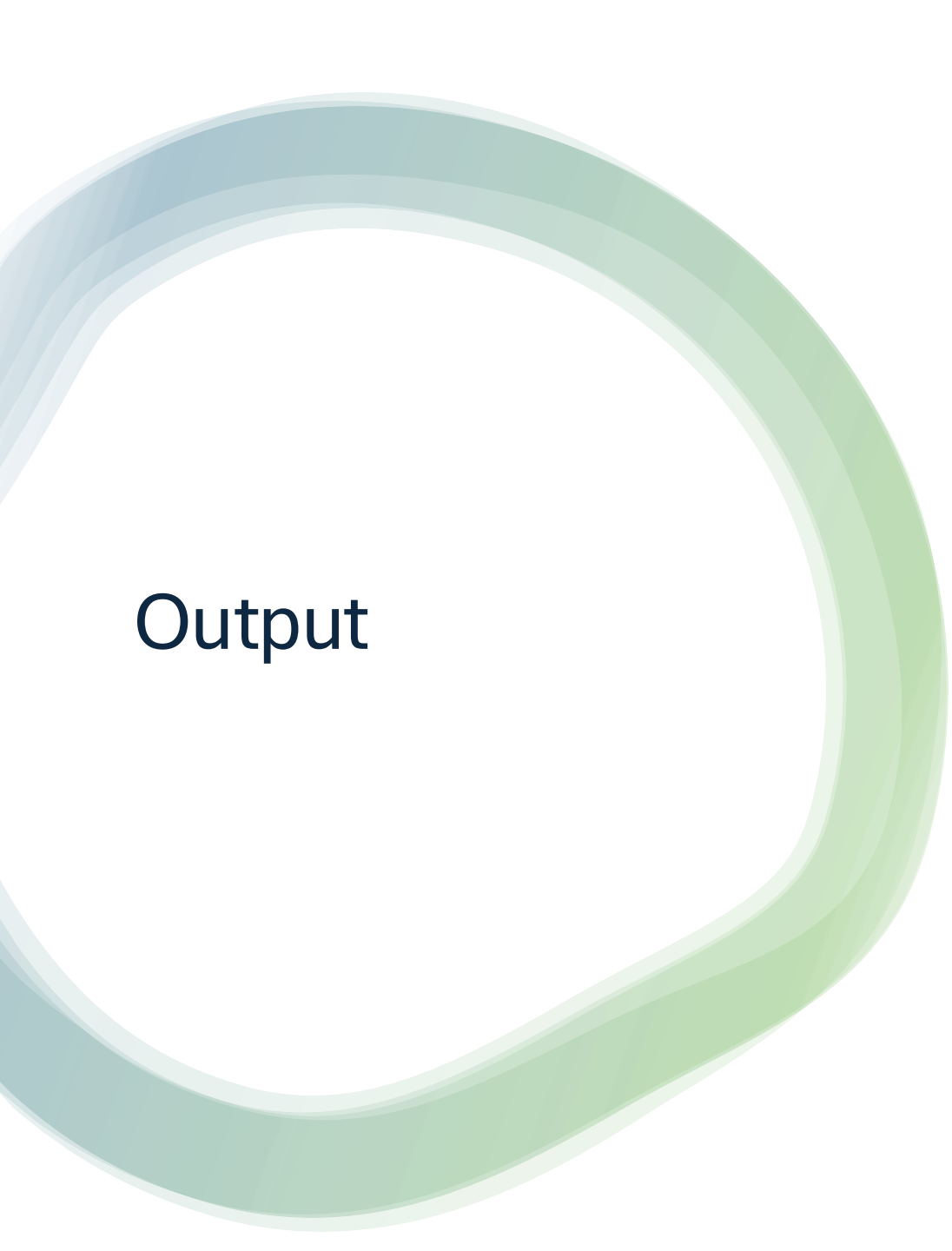
```
{ "client": "Priya Sharma", "invoice": "INV002", "tone": "Polite but Firm", "status": "Dry Run", "timestamp": "2026-05-08 23:33:18.527947" }
```

```
{ "client": "Amit Verma", "invoice": "INV003", "tone": "Formal & Serious", "status": "Dry Run", "timestamp": "2026-05-08 23:33:18.529951" }
```

```
{ "client": "TechNova Pvt Ltd", "invoice": "INV004", "tone": "Stern & Urgent", "status": "Dry Run", "timestamp": "2026-05-08 23:33:18.532948" }
```


Generated Follow-Up Emails





Output

Key Outcomes

- Automated reminders
- Personalized communication
- Faster follow-up workflow
- Better tracking
- Improved finance workflow automation

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Security & Learnings

Security Mitigations

- .env for API keys
- Dry-run email mode
- Structured templates
- Local CSV processing

Learnings

- Streamlit development
- Workflow automation
- Escalation logic
- Debugging APIs

Conclusion & Future Scope

- Future Improvements
- SMTP integration
- Database support
- AI-generated dynamic content
- Analytics dashboard

Conclusion

- Successfully built an automated finance follow-up prototype with escalation and audit tracking.
- The project successfully demonstrates AI-inspired workflow automation for finance follow-up management.





Thank You